

VPN

Virtual Private Network

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Introduction to VPN



Definition

Virtual Private Network

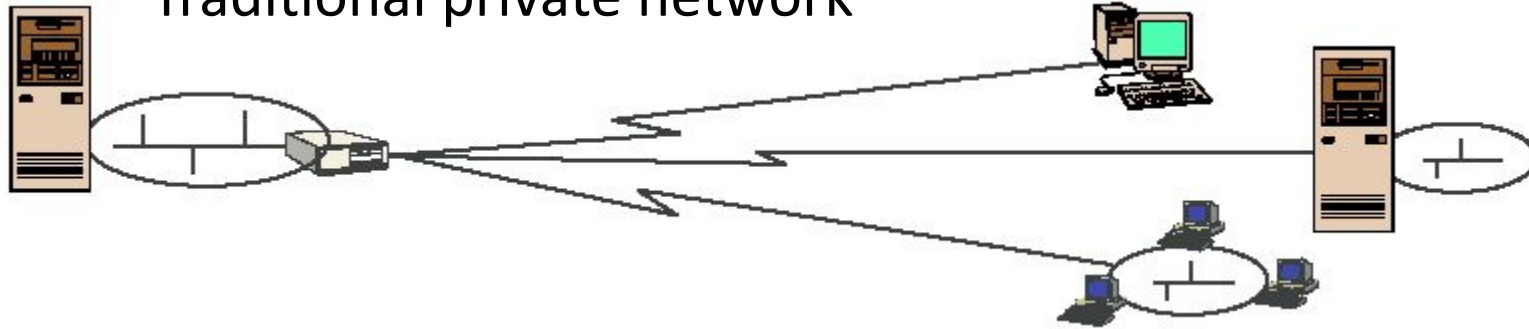
Connectivity realized on a shared infrastructure such that policies can be enforced as in a private network

- Shared infrastructure:
 - Private/public network
 - e.g., the one of an Internet Service Provider
 - IP
 - Frame Relay
 - ATM
 - The Internet
- Policies
 - Security, Quality of Service (QoS), reliability, addressing, etc.

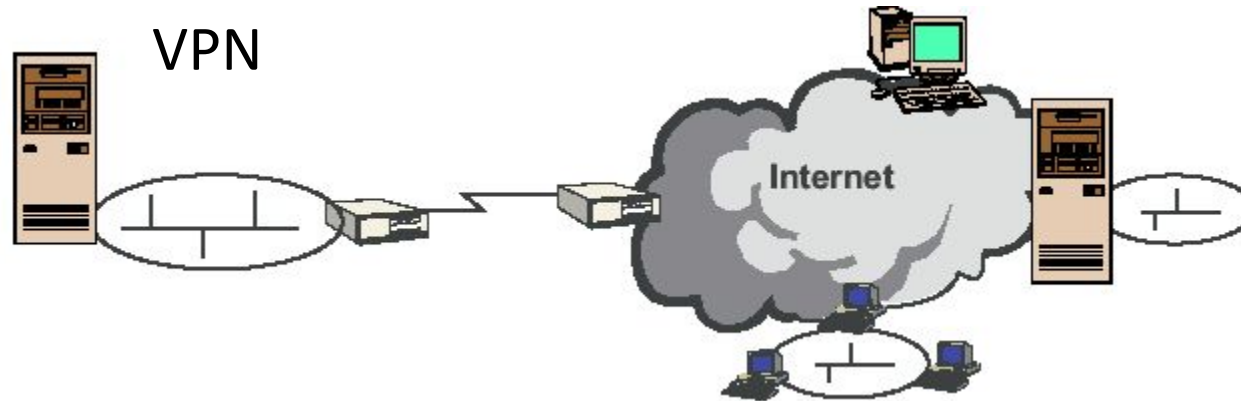
**Secure
communication**

Sample Use Case

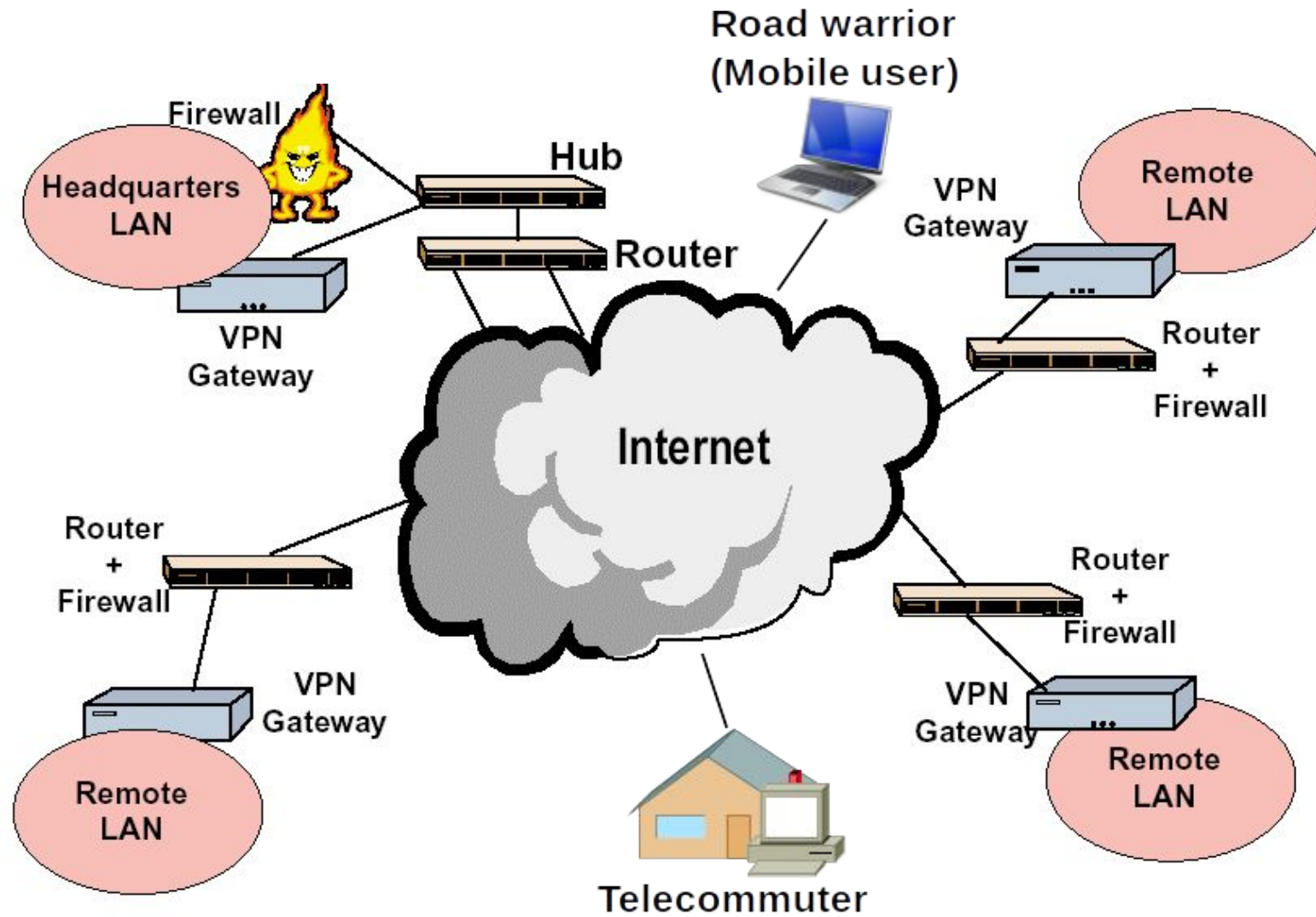
Traditional private network



VPN



Example



Key Elements

- Tunnel

- (Secure) encapsulation of corporate traffic while in transit on the shared network
- Not present in some solutions

- VPN Gateway

- Termination device on the corporate network
- Might be a tunnel endpoint

We will get back to these later on



Motivations



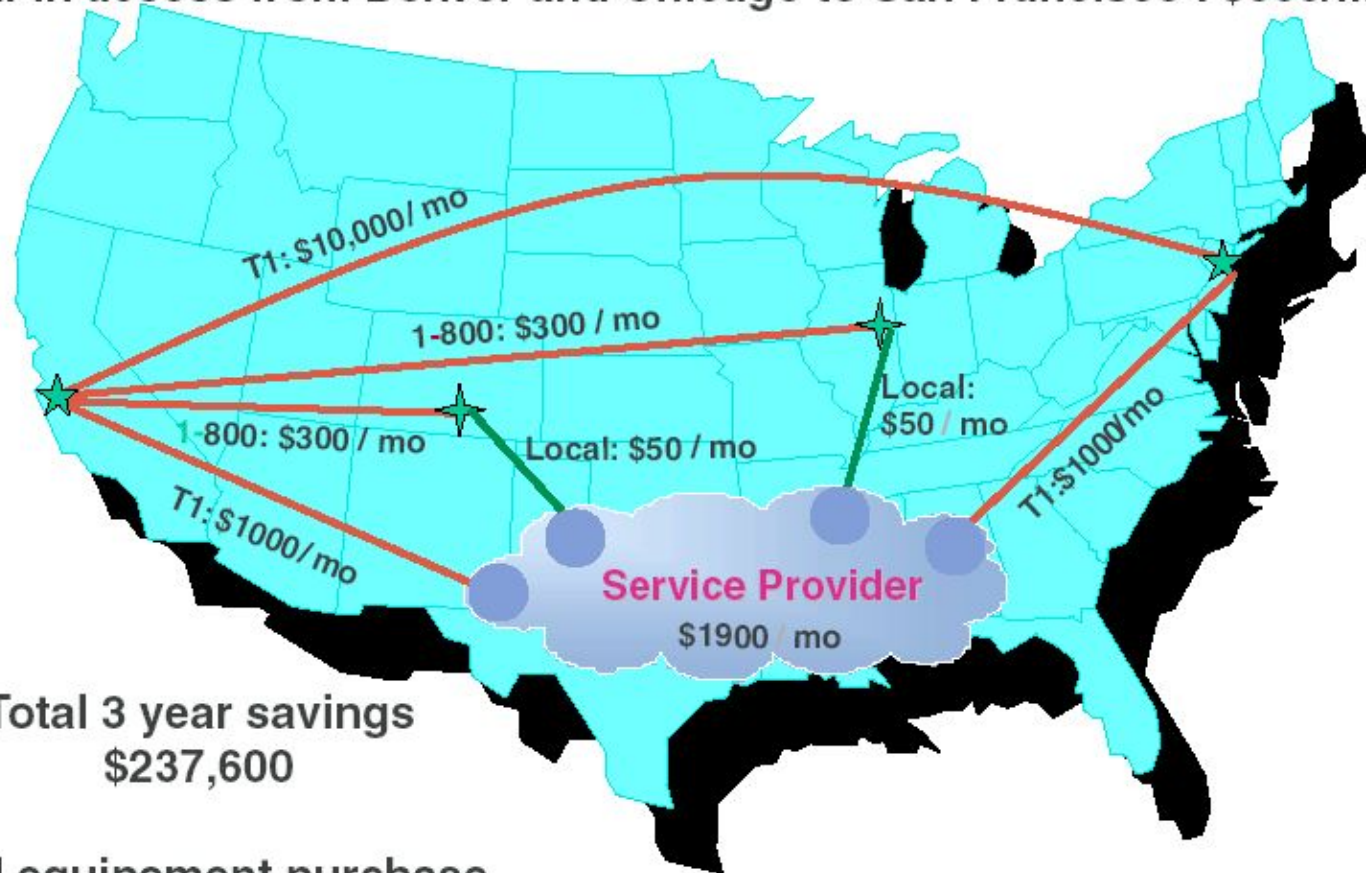
Why VPN?

- VPNs enable cutting costs with respect to expensive connectivity solutions
- Private Networks are based on
 - Private leased lines
 - Long distance dial-up solutions



An Example

T1 connections between San Francisco and New York City : \$10,000/mo
Dial-in access from Denver and Chicago to San Francisco : \$600/mo



Total 3 year savings
\$237,600

VPN equipment purchase
\$7,800

Why VPN?

- VPN enables selective and flexible access to corporate network (services)
- Limited services available to external users
 - High security
 - Few services allowed through firewall
- All intranet functionalities available to corporate users accessing from the Internet
 - VPN connection allowed through firewall
 - Services available as if connected directly to the corporate network



Basic Terminology and Scenarios



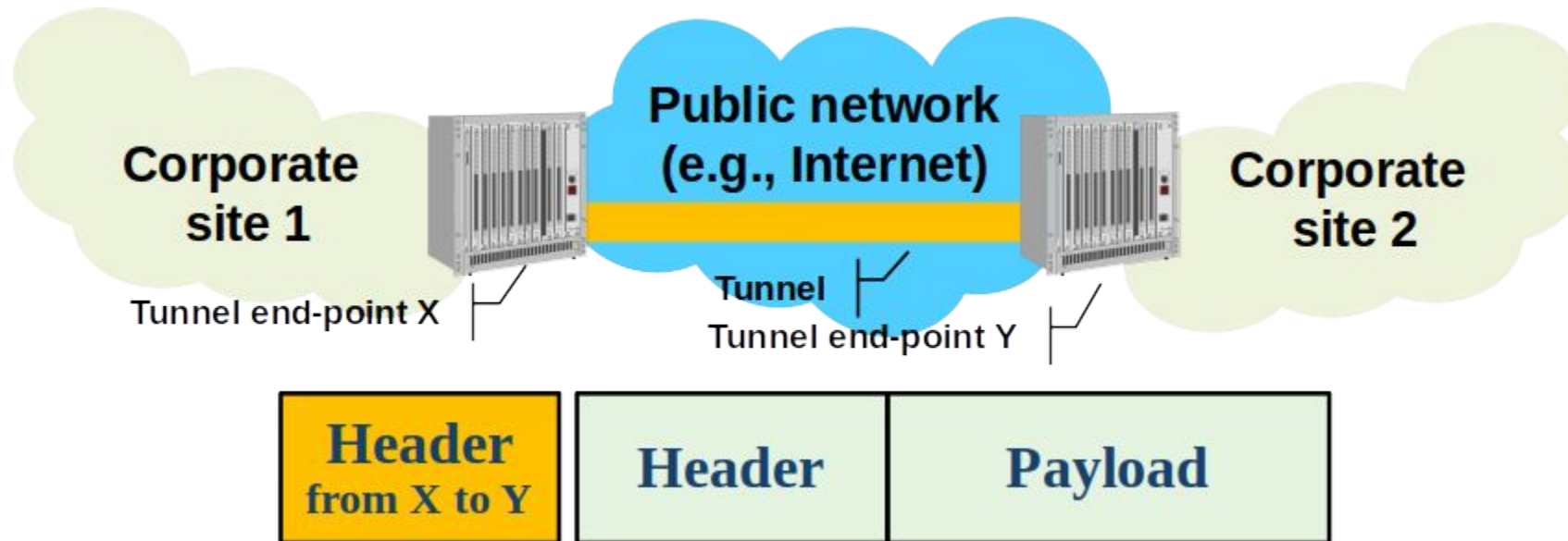
Basic Security Objectives

- End point (e.g., source/destination) authentication
 - Ensure it is what/who it declares to be
- Data integrity
 - Ensure data is not changed to prevent tampering
 - (including coming from declared source)
- Data confidentiality
 - Data cannot be accessed/read by anyone else than intended destination
- Data separation
 - Tunneling



Tunneling

- A packet (or frame) between private sites is carried through a public network within a packet handled by public nodes



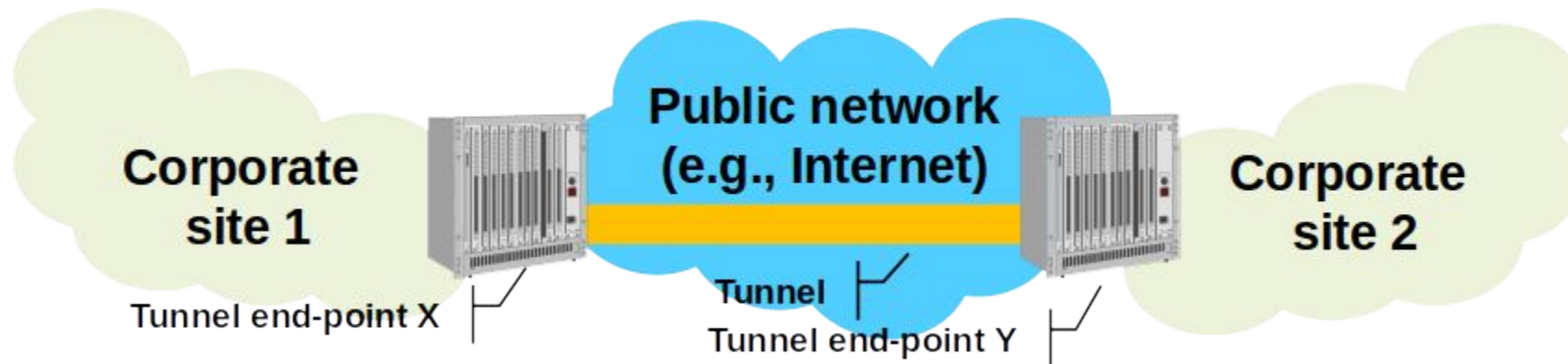
VPN Flavors

- Site-to-site VPN
 - Connect remote networks
 - Virtualizes leased line
- End-to-End VPN
 - Connect remote hosts
 - Virtualizes leased line
- Access VPN or remote VPN or virtual dial in
 - Connects terminal to remote network
 - Virtualizes (dial-up) access connection



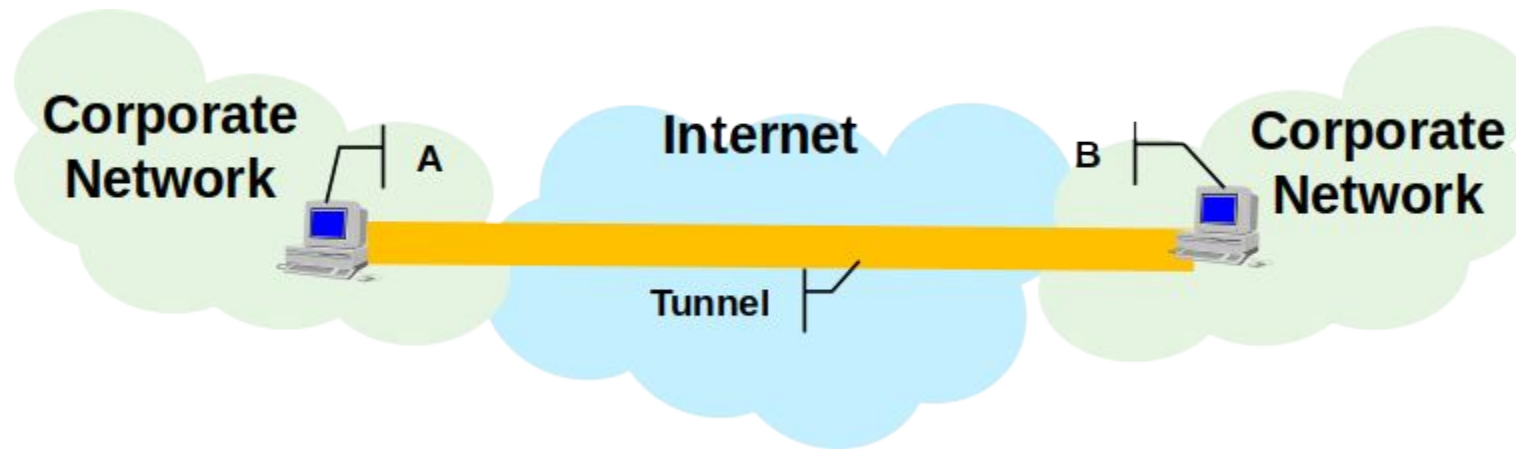
s2s VPN Tunneling

- Tunnel terminated on gateway



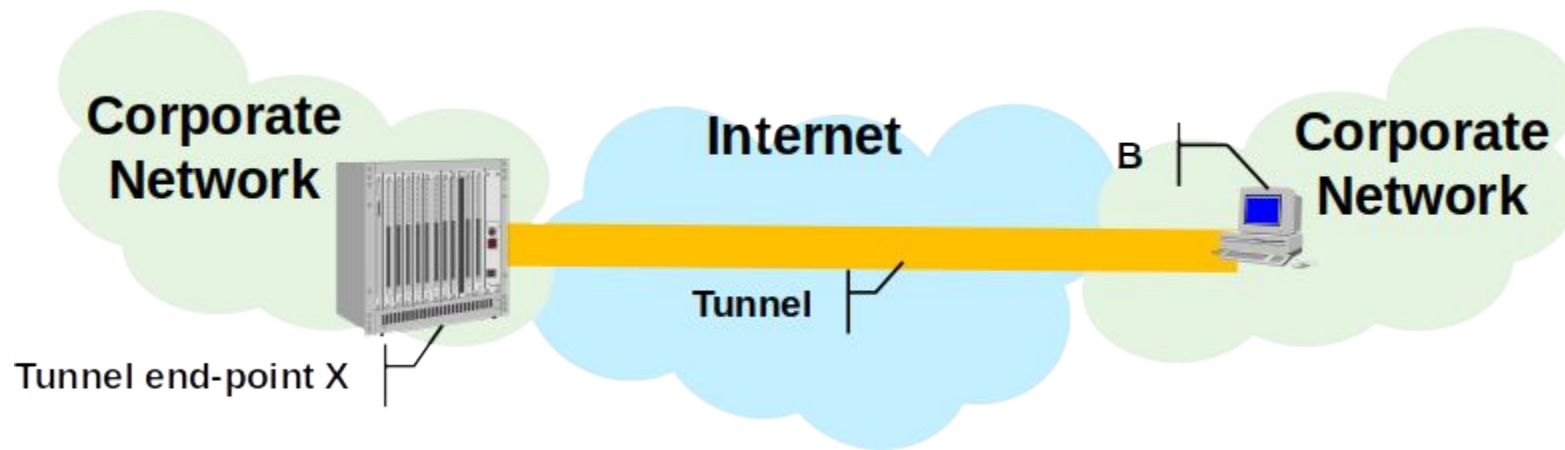
e2e VPN Tunneling

- Tunnel terminated on end systems



Remote VPN Tunneling

- Tunnel terminated on end system and vpn gateway



VPN Scenarios

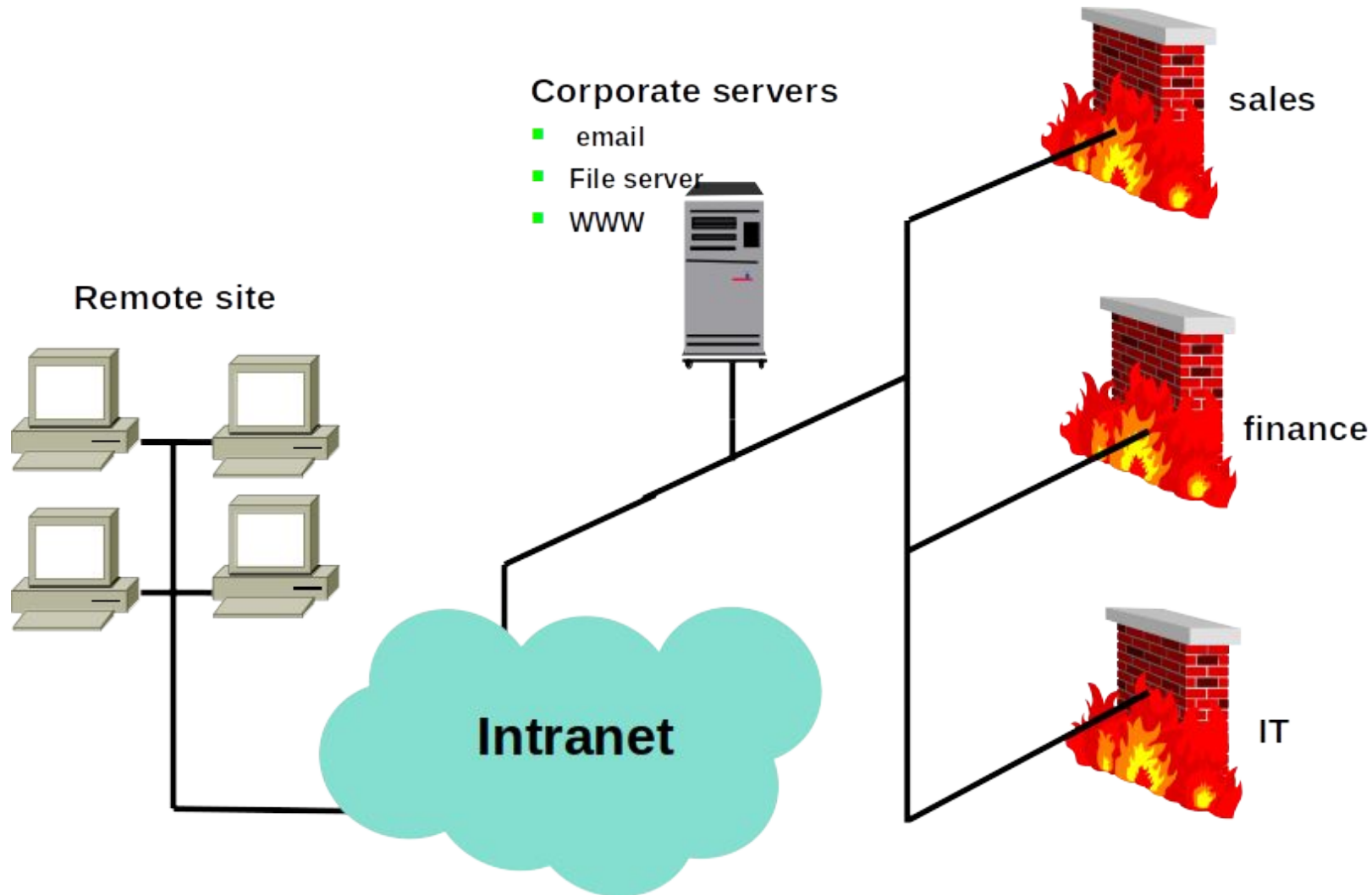
- Intranet VPN

- Interconnection of corporate headquarters, remote offices, branch offices, telecommuter, traveling employee

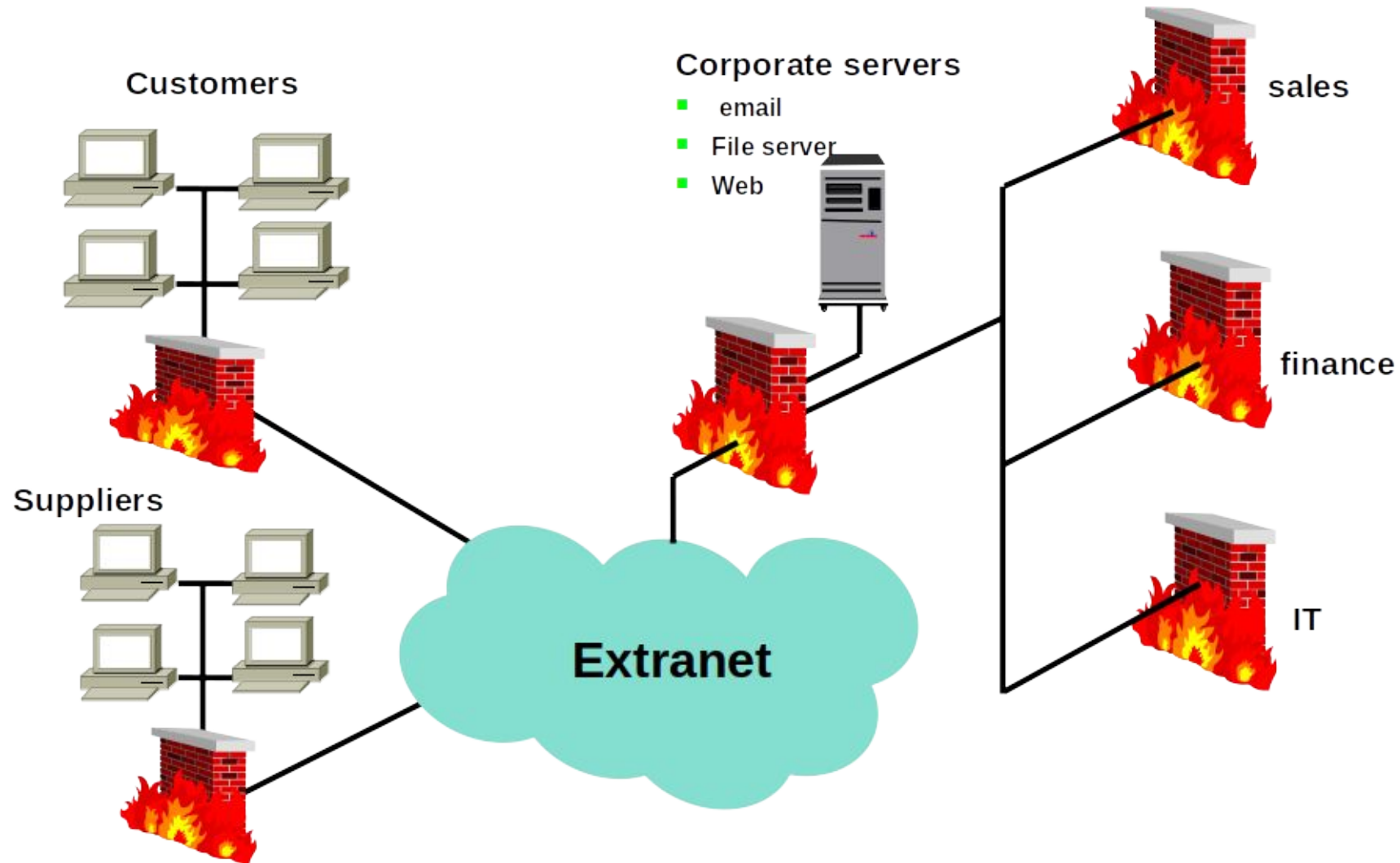
- Extranet VPN

- Interconnection of customers, suppliers, partners, or communities of interest to a corporate intranet
- Provide controlled access to an individual customer/partner/provider user

Sample Intranet Architecture



Sample Extranet Architecture



Extranet Specific Issues

- Restricted access to network resources from interconnected networks
 - Firewall at the VPN
- Overlapping Address Spaces
 - Network address translation
- Open, standard-based solution
 - Enables interoperability among different organizations
- Traffic control
 - Avoid that partner traffic compromises performance on corporate network



Internet Access

■ Centralized

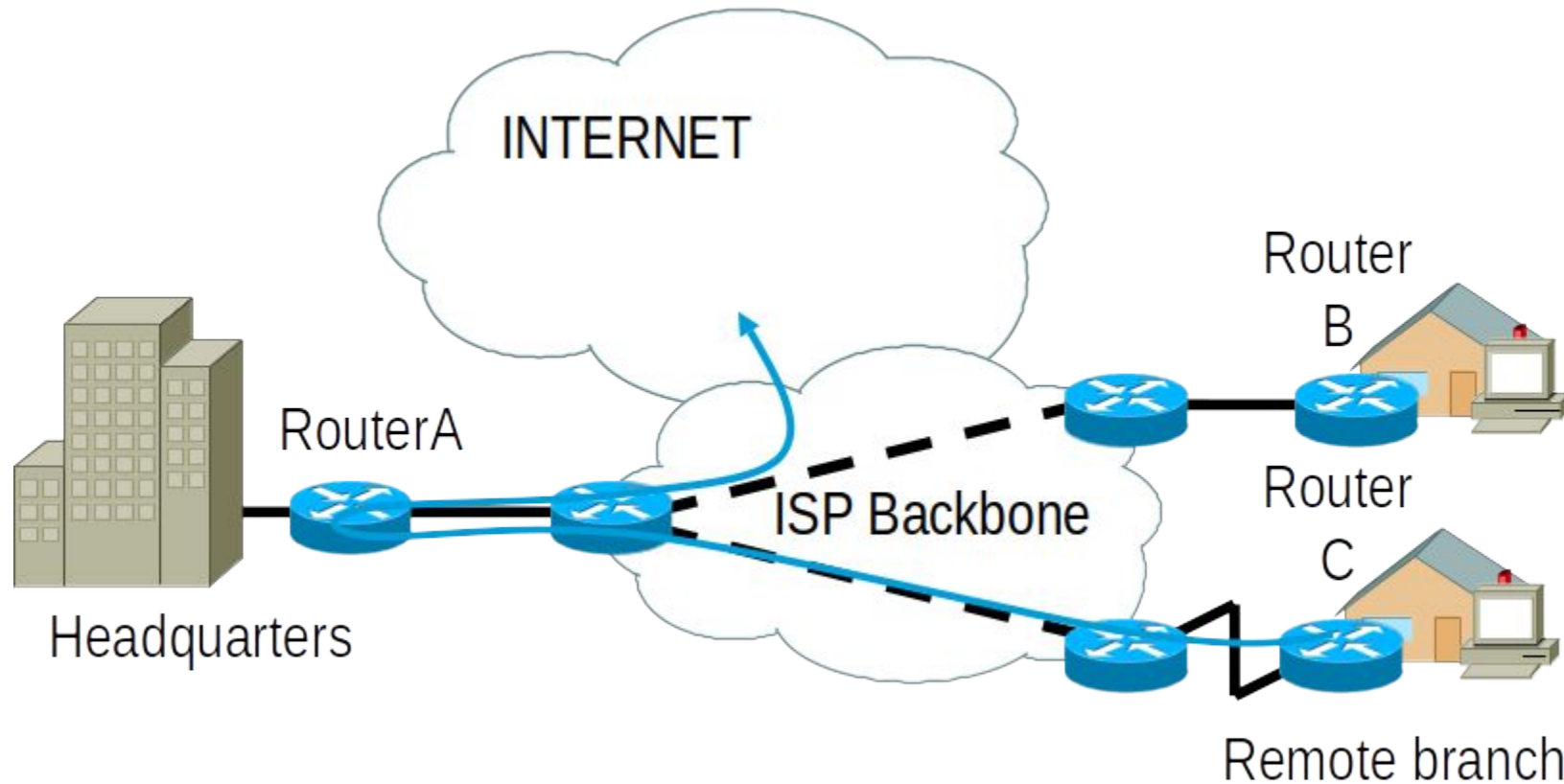
- Remote branches/users use public IP network only to reach headquarters
- Internet access only from headquarters
- VPN carries also traffic to and from the Internet
- Centralized access control
 - Firewall

■ Distributed (voluntary connection)

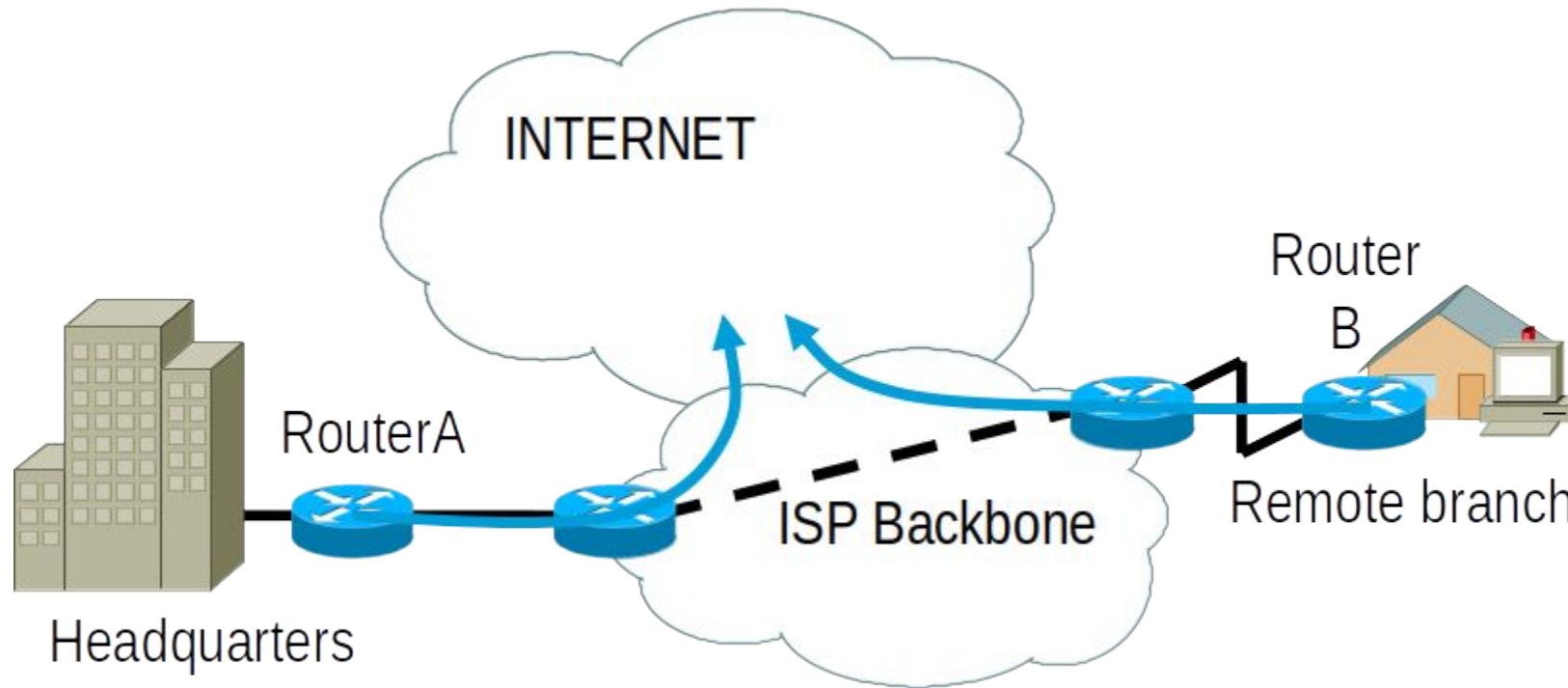
- Remote branches/users access the Internet through their IP network connection
- VPN is deployed only for corporate traffic



Centralized Internet Access



Distributed Internet Access



(Virtual) VPN Topologies

■ Hub and spoke

- Each branch communicates directly with headquarters
- Fits to data flow of many corporations
 - Mainframe or data-center centered
- Routing is sub-optimal
- Small number of tunnels
- Hub could become bottleneck

■ Mesh

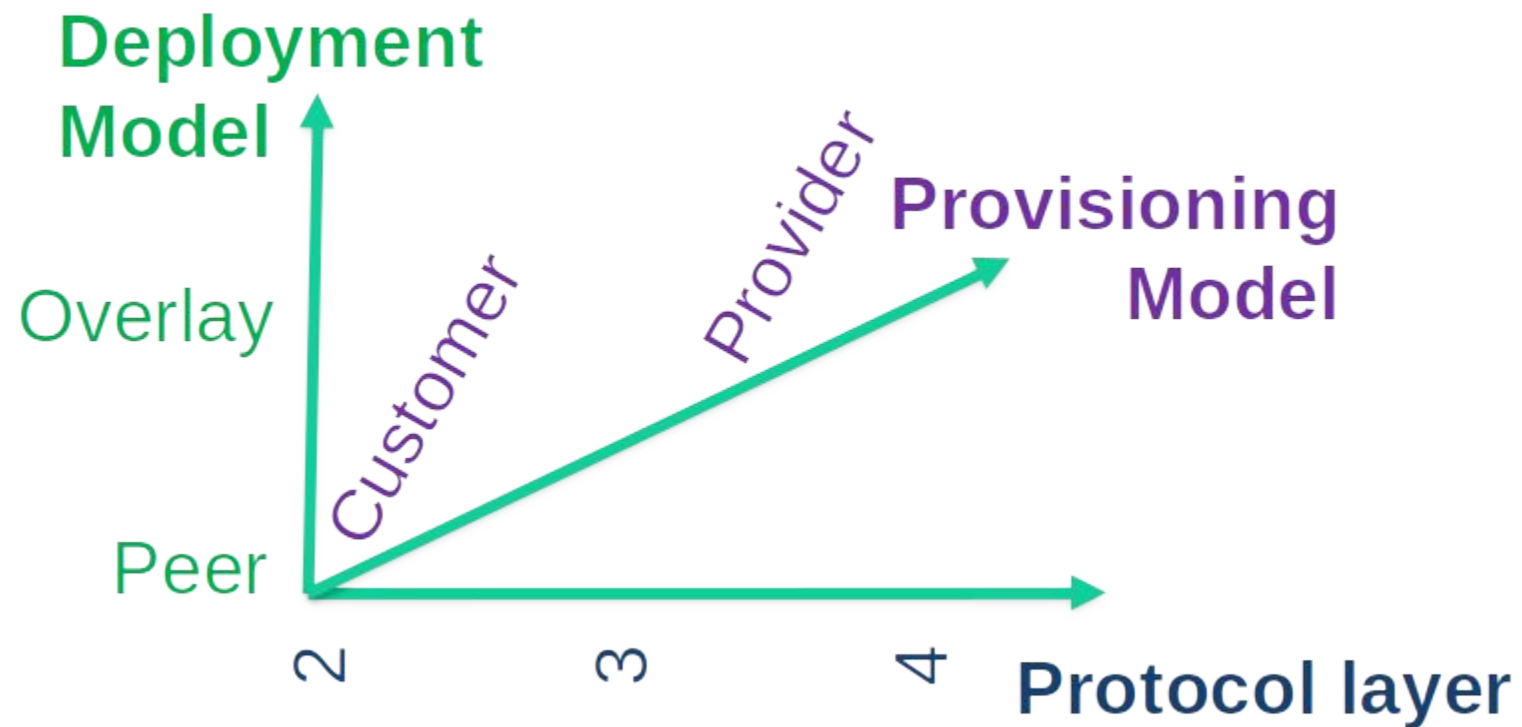
- Larger number of tunnels
 - Harder to manually configure
- Optimized routing

Deployment Models, Provisioning Models & VPN Protocol Layers

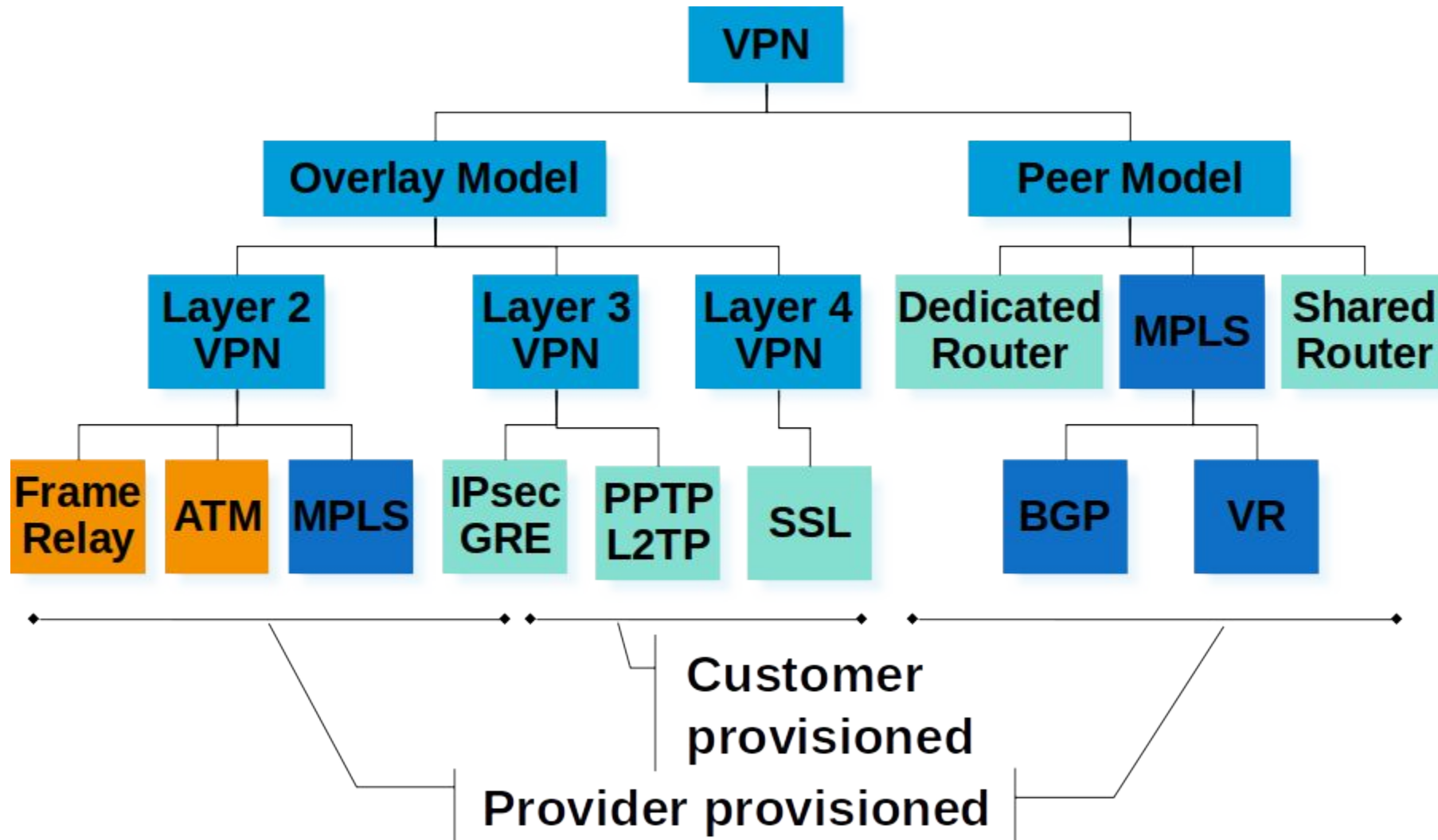


Many VPN solutions

- Three dimensions



VPN categories



Overlay Model

- The public network does not participate in realizing the VPN
 - It does not know where VPN destinations are
 - Just connectivity among VPN gateways
- Each VPN gateway must be “in touch” with every other VPN gateway
 - E.g., highly meshed tunnels
- Routing is performed by the VPN gateways



Peer Model

- Each VPN gateway interacts with a public router (its peer)
 - Exchange of routing information
 - Service provider network disseminates routing information
- Public network routes traffic between gateways of the same VPN

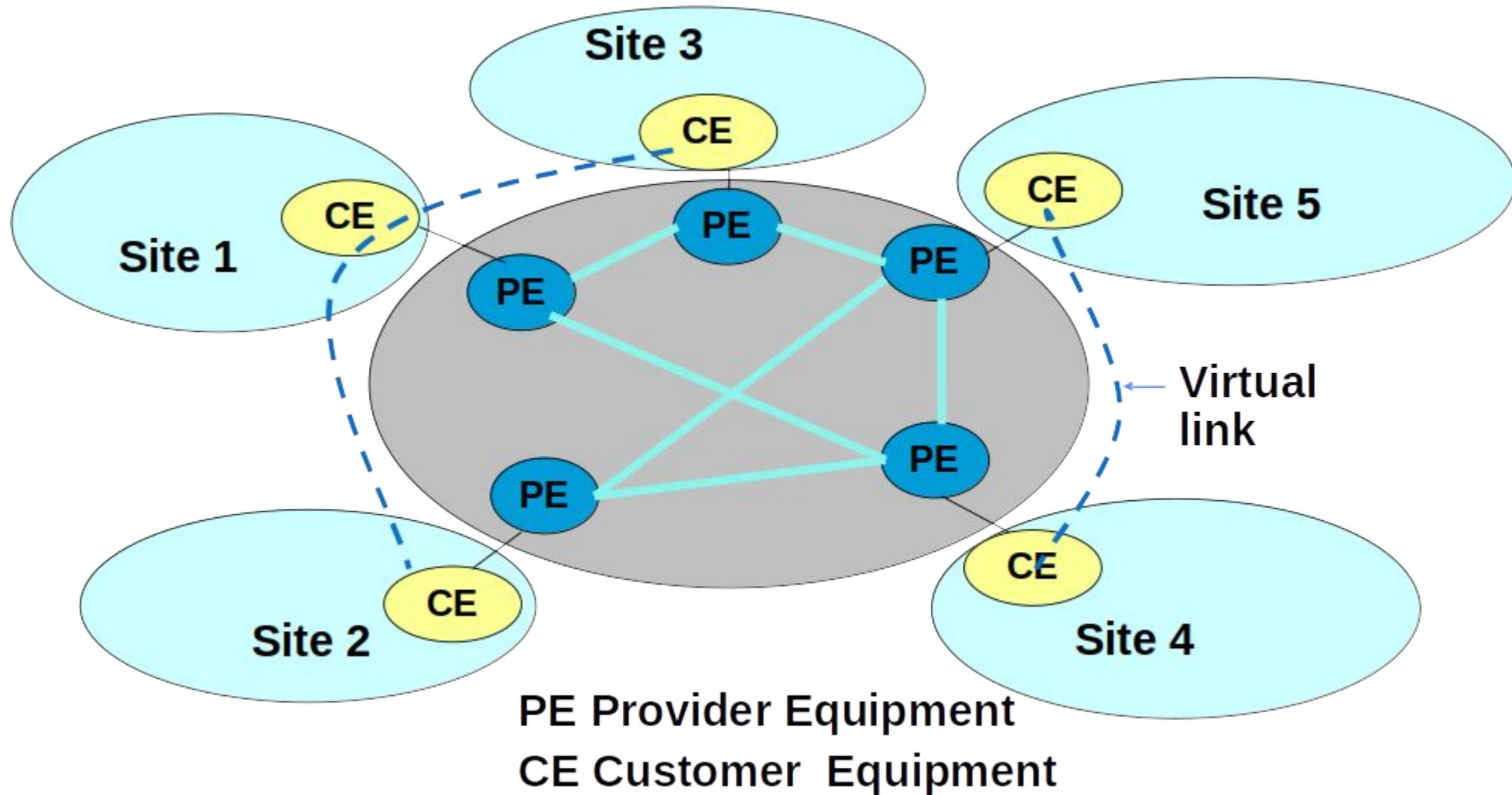


Customer Provisioned VPN

- Customer implements VPN solution
 - Owns, configures, manages devices implementing VPN functionalities
 - Customer equipment
- Network provider is not aware that the traffic generated by customer is VPN
- All VPN features implemented in customer devices
- CE terminates tunnels



Customer Provisioned VPN

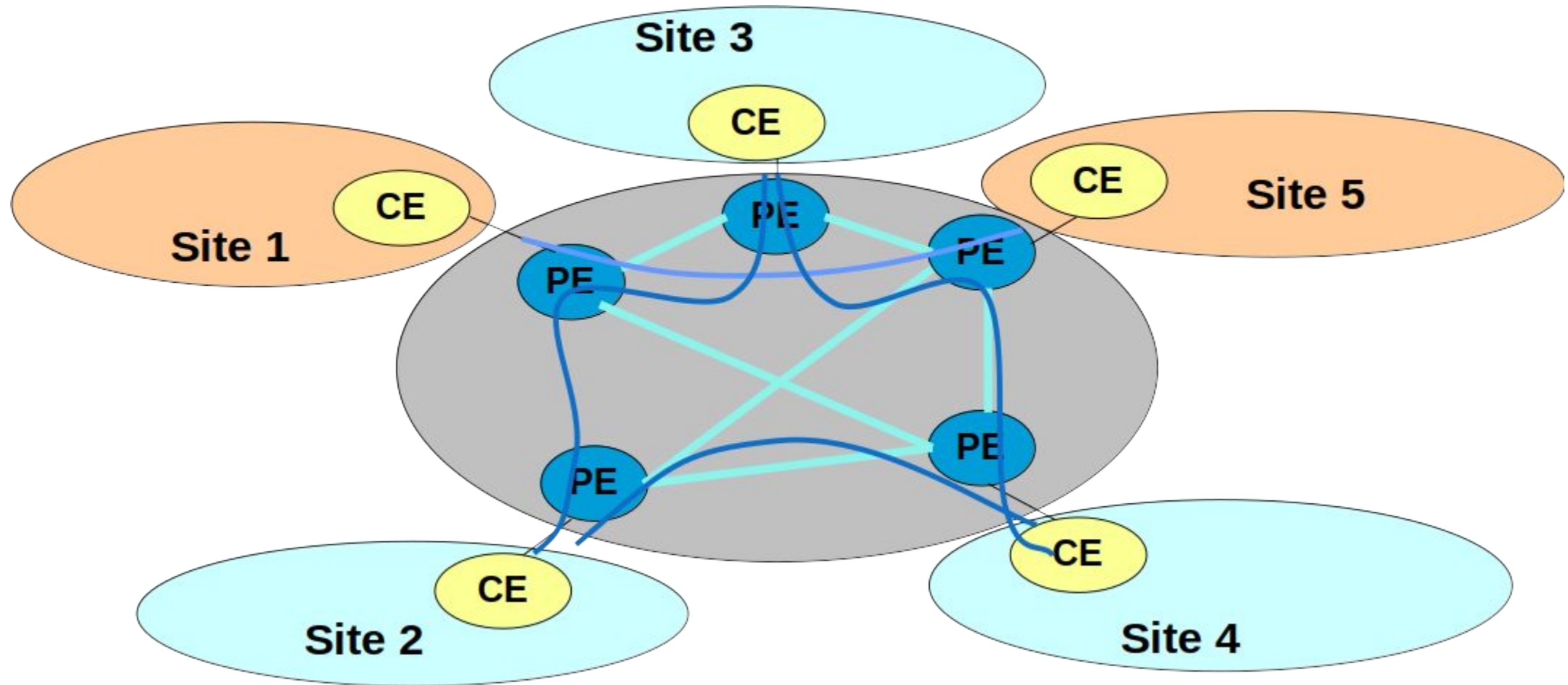


Provider Provisioned VPN

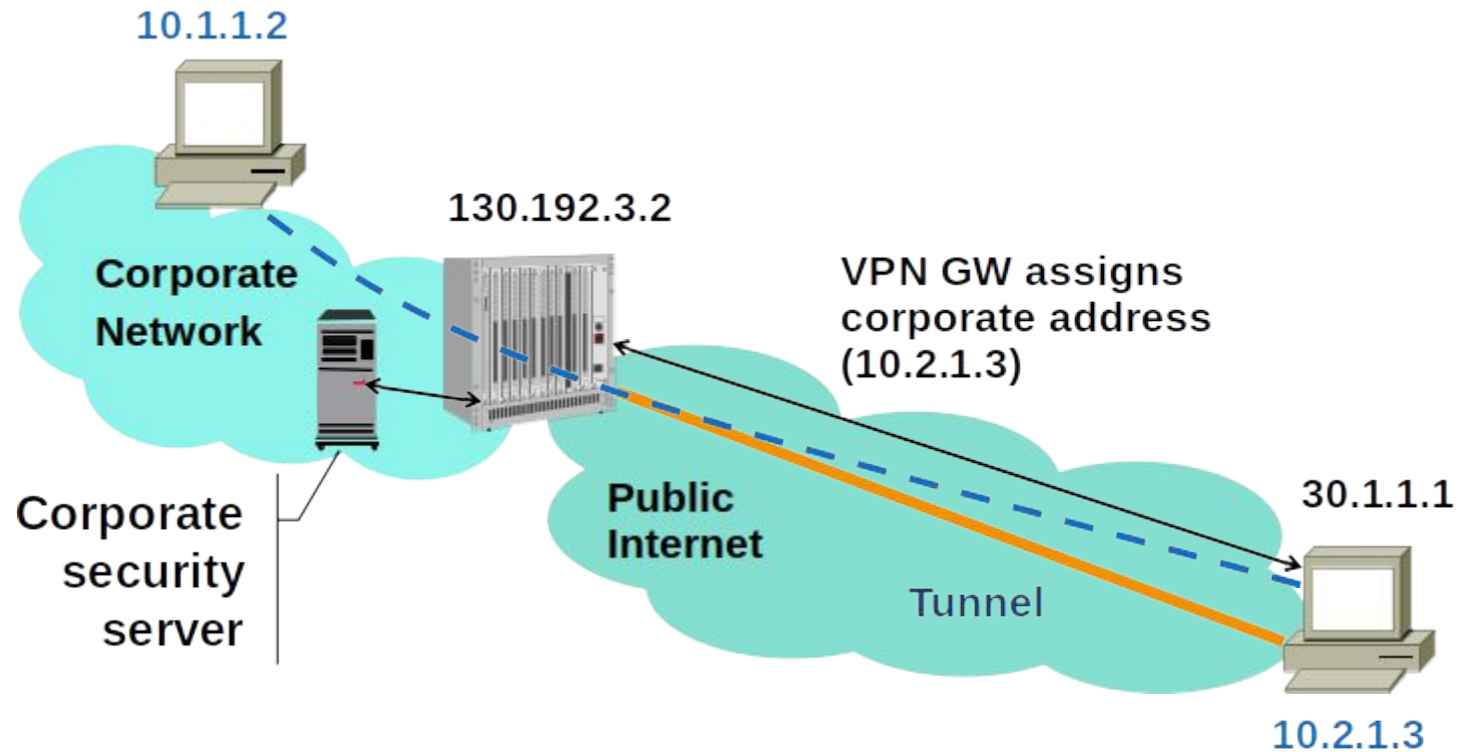
- Provider implements VPN solution
 - Owns, configures, manages devices implementing VPN functionalities
- VPN state maintained by the provider devices
- Traffic belonging to different VPNs is separated by the provider devices
- CE may behave as if it were connected to a private network
- PE terminates tunnels



Provider Provisioned VPN



Access VPN Customer Provisioned



Tunnel endpoints: 130.192.3.2 – 30.1.1.1

Connection endpoints: 10.2.1.3 – 10.1.1.2

Customer VS Provider Provisioned

■ Customer Provisioned

- Remote host has 2 addresses
 - ISP assigned and corporate
- Remote host terminates VPN tunnel
- Remote host must activate tunnel
 - If tunnel is not activated, client can operate without VPN
- Can be used from any Internet connection (ISP)

■ Provider provisioned

- Remote host has 1 address (corporate)
- NAS terminates VPN tunnel
- Remote host is always on VPN
- Internet access is only centralized
- Requires access to specific ISP



Layer N VPN

- Packet transport (tunneling) provided:
 - by Layer N Protocol and/or
 - as Layer N service

Layer 2 VPNs

■ Virtual Private LAN Service

- Emulates functionalities of LANs
- Can be used to connect LAN segments
 - Works as single LAN through the public network
- VPN solution emulates learning bridges
 - Routing based on MAC addresses

■ Virtual Private Wire Service

- Emulates a leased line
- Any protocol can be carried

■ IP-Only LAN-like Service

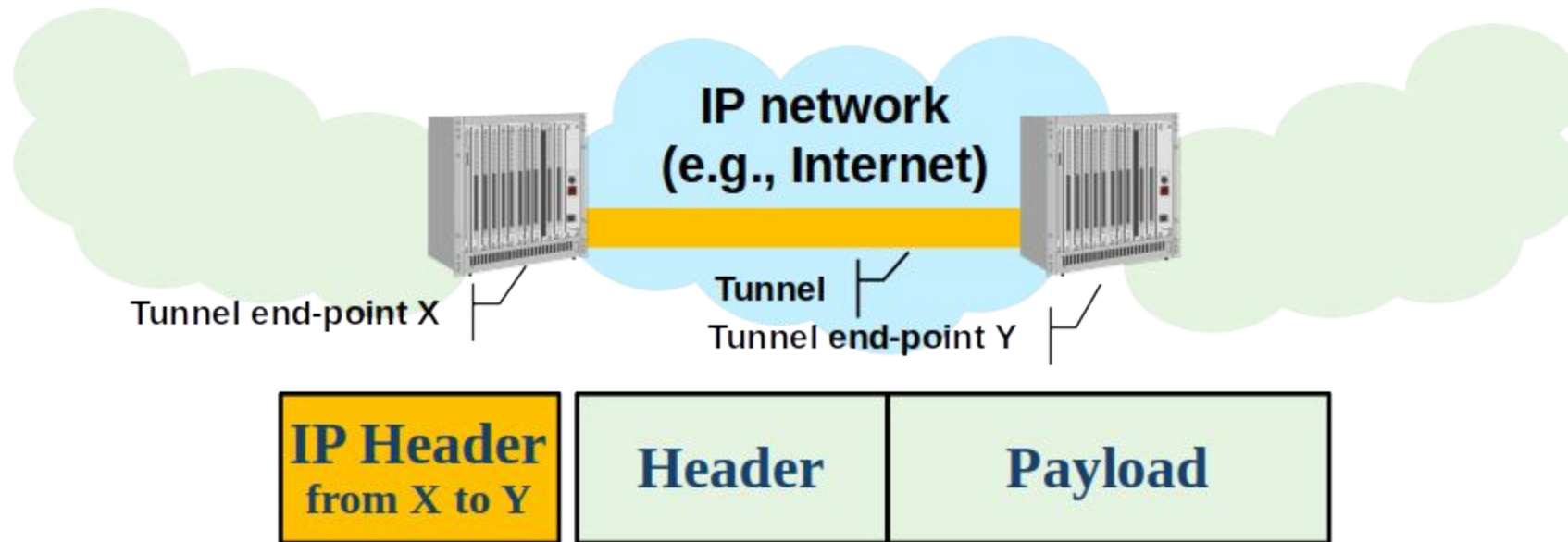
- CEs are IP routers or IP hosts (not Ethernet switches)
- Only IP (plus ICMP and ARP) packets travel through the VPN

Layer 3 VPNs

- Layer 3 packets are forwarded through the public network
- Routing based on layer 3 addresses
 - Peer: VPN/corporate/customer addresses
 - Overlay: backbone addresses
- CEs are either IP routers or IP hosts

Tunneling in Layer 3 VPN

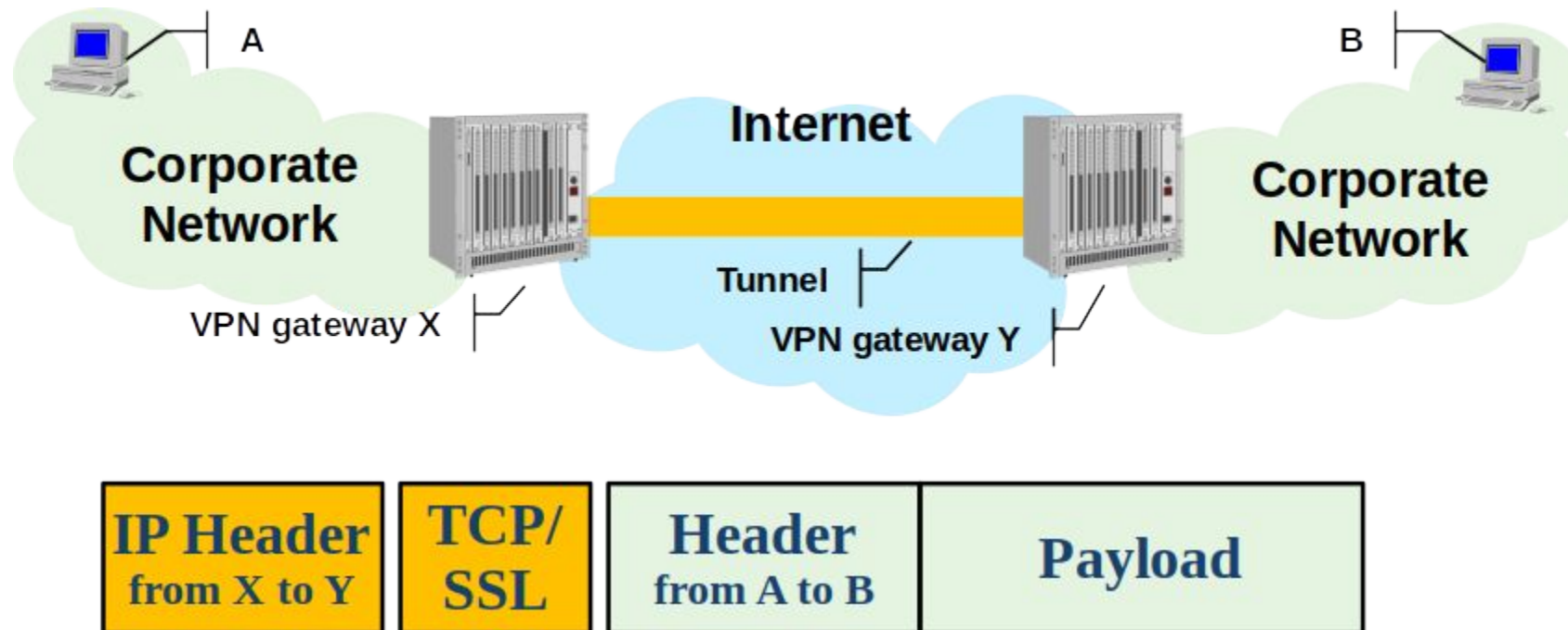
- A packet (or frame) is carried through an IP network within an IP packet



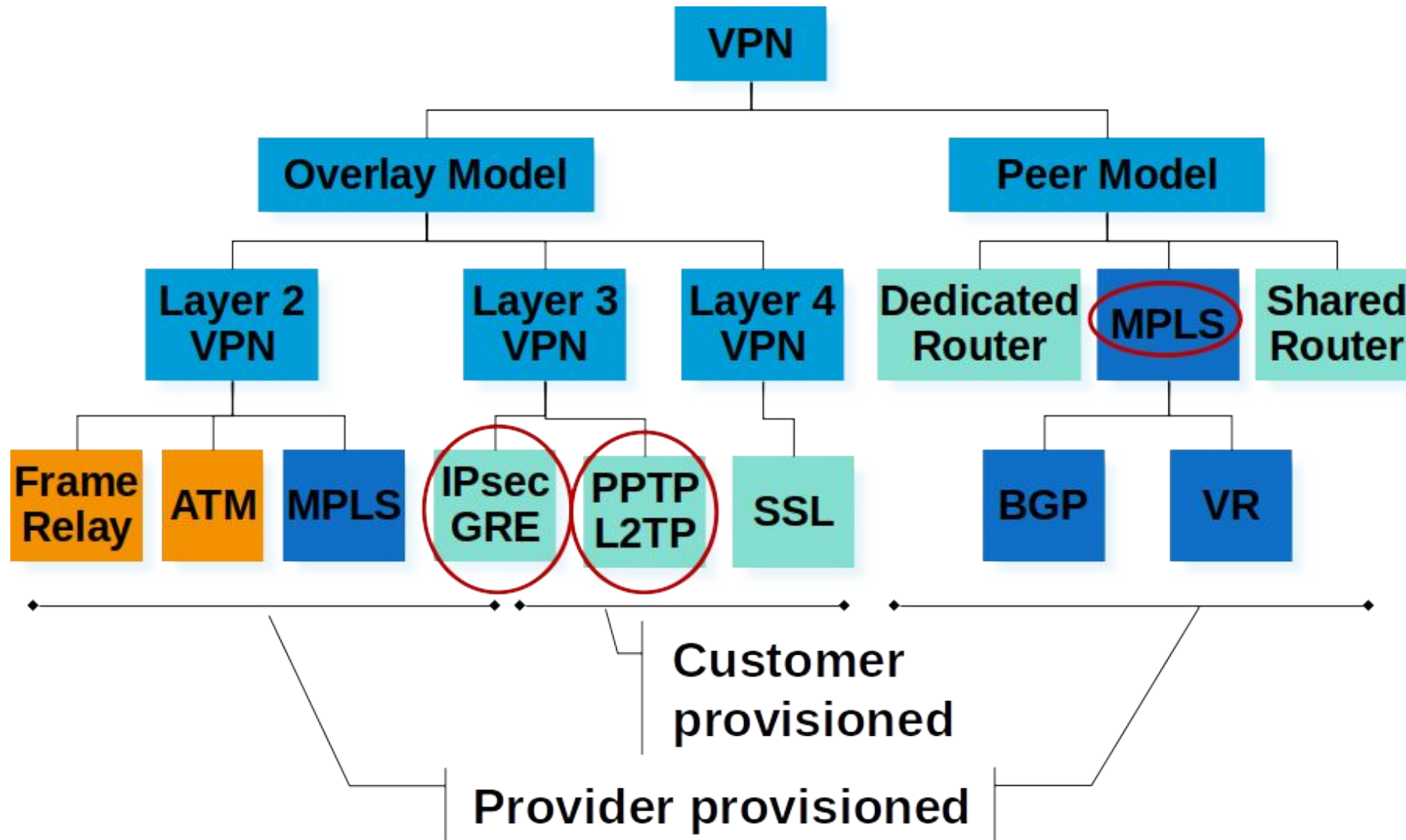
- An IP packet within an IP packet (IP-in-IP)
 - GRE, IPsec
- A layer 2 frame, within an IP packet
 - L2TP, PPTP (based on GRE)

Layer 4 VPN Tunneling: s2s

- VPN built using TCP connections
 - Tunnels realized by TCP connections
- Security achieved with SSL/TLS



Many VPN solutions

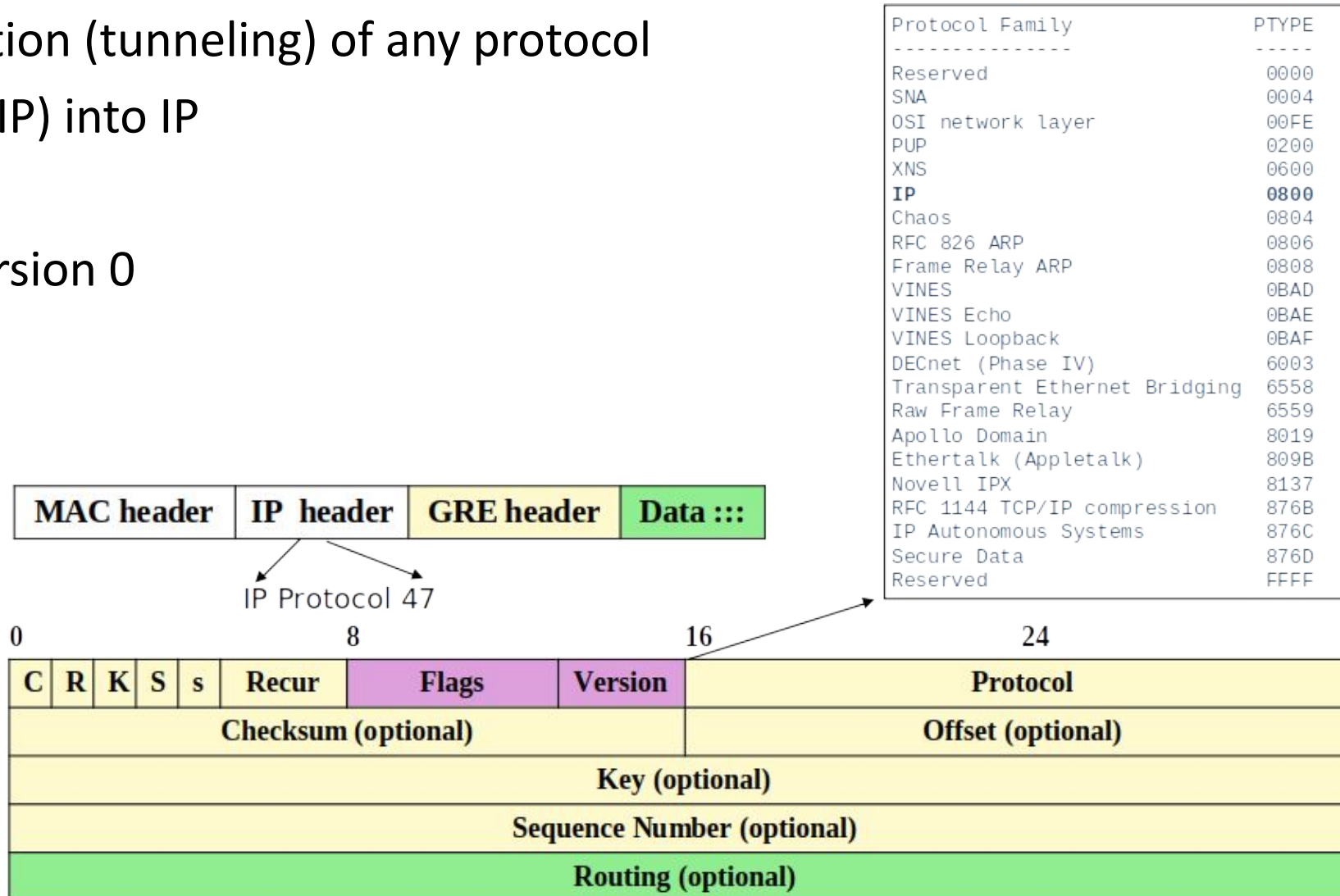


GRE – Generic Routing Encapsulation



Packet Format

- Encapsulation (tunneling) of any protocol (including IP) into IP
- Header version 0



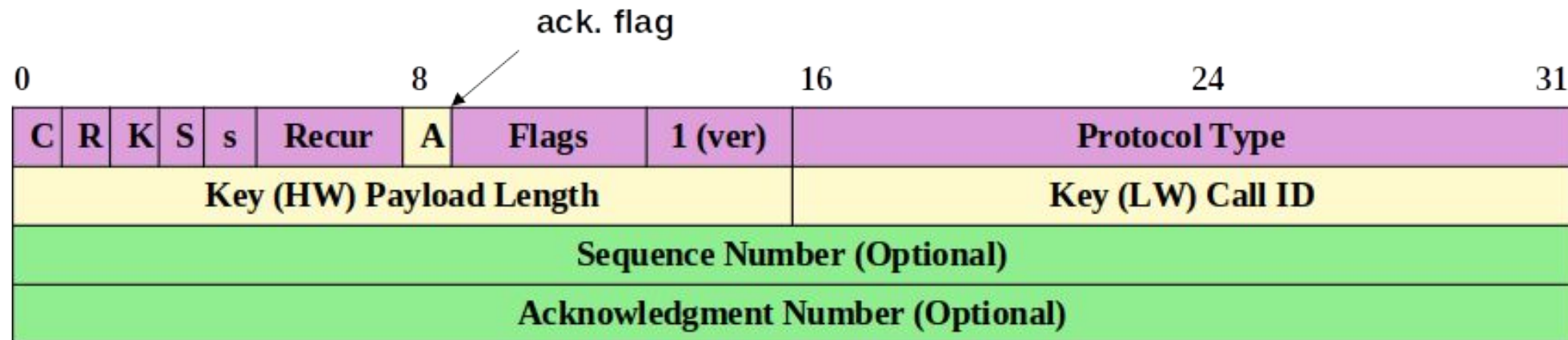
Header fields

- C, R, K, S
 - Flags indicating the presence/absence of optional fields
- S
 - Strict source routing flag
 - if the destination is not reached when the source route list end, the packet is dropped
- Recur
 - Max. number of additional encapsulation permitted (must be 0)
- Protocol
 - ID of the payload protocol
- Routing
 - Sequence of router IP addresses or ASs for source routing



Enhanced GRE (version 1)

- Deployed by PPTP
- Acknowledgment Number
 - Delivery of packets by remote end-point can be notified



Advanced Functionalities

- Key (high 16 bit)
 - Payload length: no. of bytes excluding GRE header
- Key (low 16 bit)
 - Call ID: session ID for this packet
- Sequence number
 - For ordered delivery, error detection and correction
- Acknowledge number
 - Highest number of GRE packet received in sequence for this session
 - Cumulative ack



Other mechanisms in GRE

- Flow control
 - Sliding window mechanism
- Out-of-order packets
 - Discarded, because PPP allows lost packets, but cannot handle out-of-order packets
- Timeout values
 - Re-computed each time an ack packet is received
- Congestion control
 - Timeouts do not cause re-transmission
 - Used only to move sliding window
 - Packets will be lost
 - Their value should be rapidly increased



**Layer 2 frame, within an IP
packet**



Two Protocols for Access VPN

- L2TP (Layer 2 Tunneling Protocol)
 - Initially only Provider provisioner
 - (Initially) not widely implemented in terminals
 - Independent of layer 2 protocol on host
 - Security through IPsec
 - Strong, but complicated
- PPTP (Point-to-Point Tunneling Protocol)
 - Customer provisioner
 - Originally proposed by Microsoft, Apple, ...
 - Weak encryption and authentication
 - Proprietary key management



L2TP - Layer 2 Tunneling Protocol

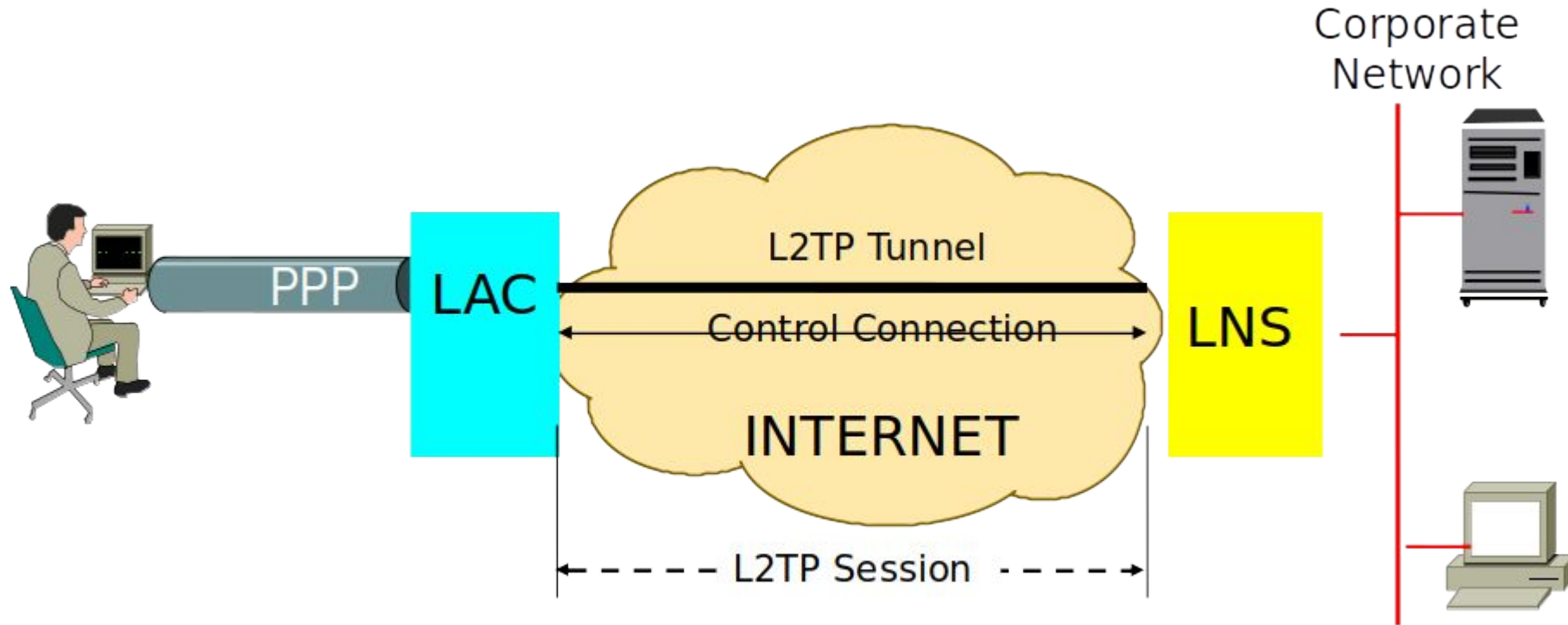


Solution Components

- Tunneling between public network access point and corporate network
 - Also wholesale dial-up services
 - Between access provider and Internet Service Provider
- L2TP Access Concentrator (LAC)
 - Network access device supporting L2TP
 - NAS (Network Access Server)
- L2TP Network Server (LNS)
 - Corporate (VPN) Gateway
- Customer provisioned deployment mode by including LAC functionality in host



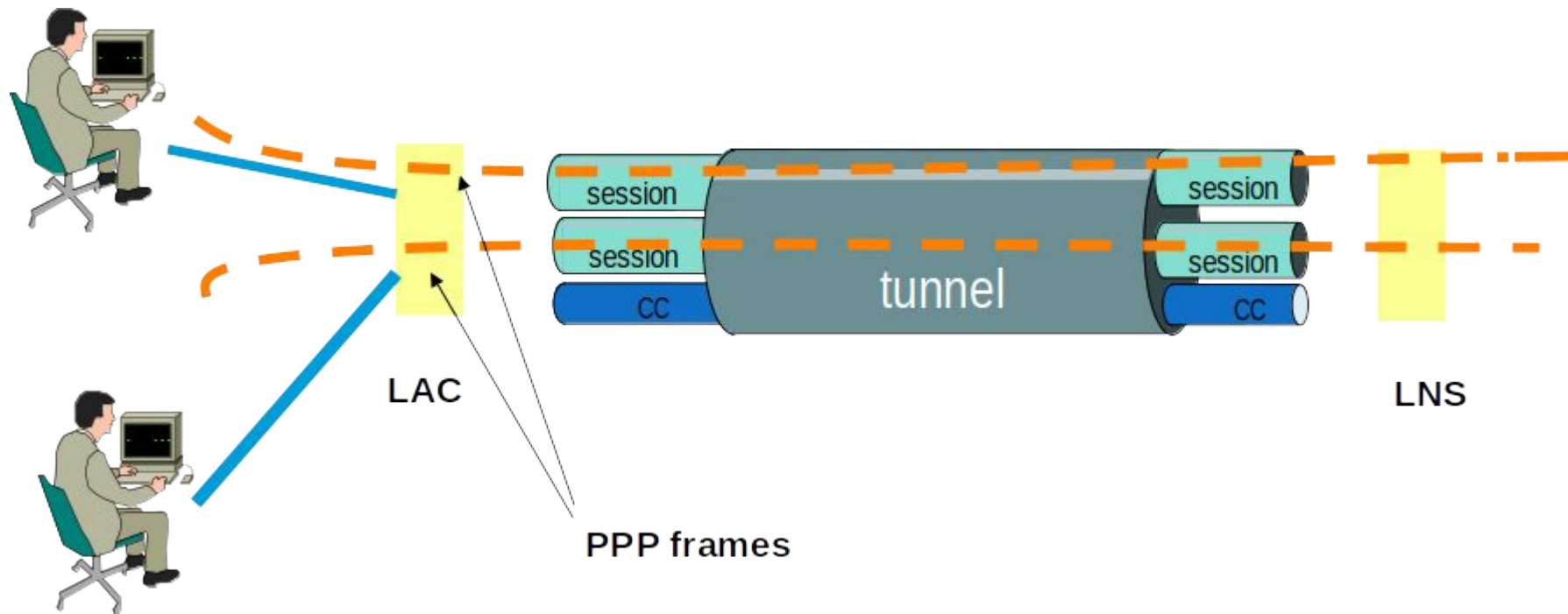
Original Reference Scenario



Provider provisioned deployment mode

Tunnels and sessions

- Multiple sessions may exist within the same tunnel
- Multiple tunnels may be established between the same LAC and LNS or multiple LNSs



L2TP Operation

- Establish a **control connection** for a tunnel between LAC and LNS
- Establish one or more **sessions** triggered by a call request
- The control connection **must** be established before a connection request is generated
- A session **must** be established before tunnelling PPP frames

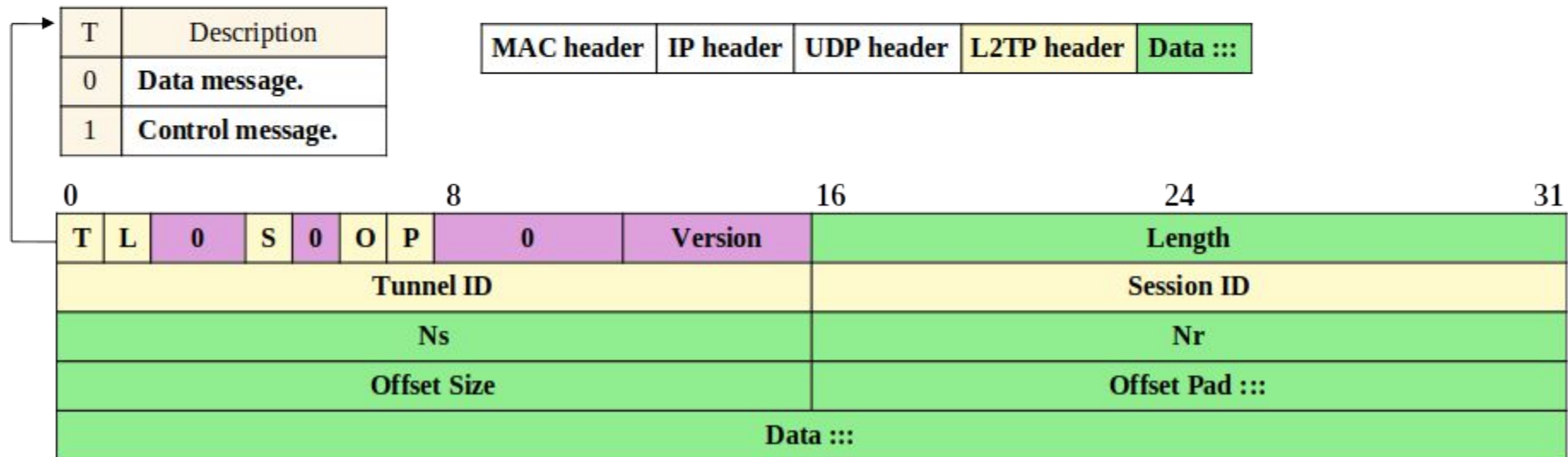
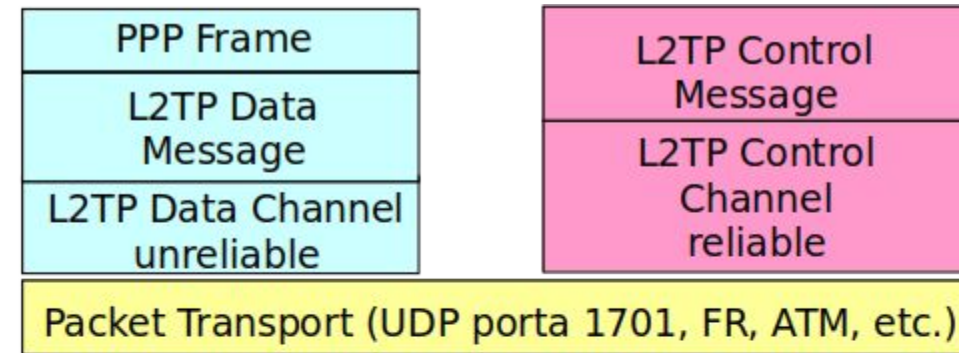


Establishing a tunnel

- Peer can be authenticated
- A shared secret must exist between LAC and LNS
- L2TP uses a CHAP-like mechanism
 - Challenge-Handshake Authentication Protocol
 - A challenge is proposed to the other peer
 - The correct answer to the challenge requires the shared secret
 - Cryptographic algorithm used to create the answer
- The tunnel endpoints exchange the local ID attributed to the tunnel

L2TP Header

- Control Message
- Data Message



Header fields

- L, S, O
 - Flags indicating whether the fields length, Ns & Nr and offset are present
 - For control messages L=S=1 and O=0
- P
 - Priority flag, if set, the priority is high
- Ver
 - Version, must be 2
- Tunnel ID
 - Recipient's ID of the control connection (local meaning)
- Session ID
 - Recipient's ID of the session within the same tunnel (local meaning)

Other header fields

- **Ns**
 - Sequence number of the data or control message
- **Nr**
 - Sequence number of the next control message to be received (i.e. last Ns received in order +1 modulus 216)
- **Offset**
 - Number of bytes, past the header, where the payload data starts

Sequence Numbers

- Data connections use sequence numbers only to detect out of order packets
- No re-transmission for data streams
- No ack in data streams
 - Layer 2 protocol, e.g., HDLC, can possibly take care of this
- Control packets use ack and retransmission
 - Selective repeat
 - Tx and Rx windows set to 32k



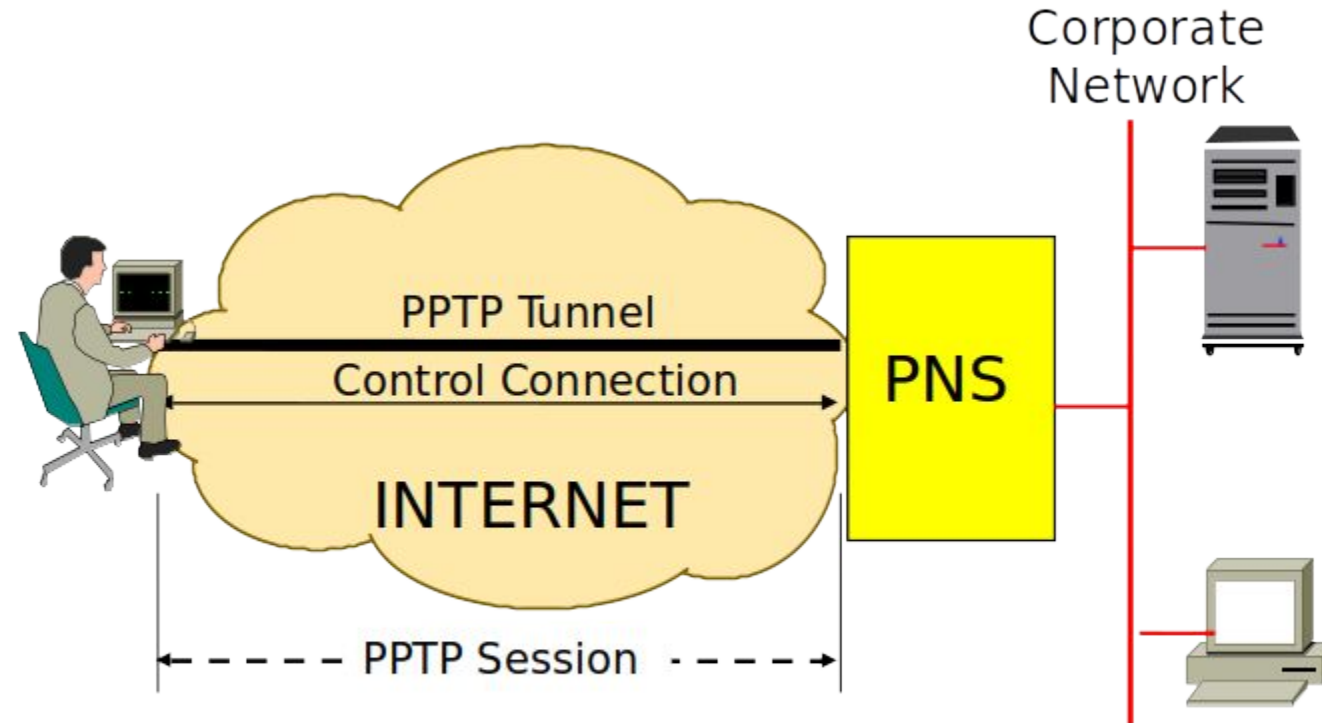
Security issues

- Tunnel endpoint authentication
 - Authentication only during tunnel establishment phase
 - A user who can snoop traffic, can easily inject packets in a session
 - Tunnel and session IDs should be selected in an unpredictable way (not sequentially)
- Packet level security
 - Encryption, authentication, and integrity **must be provided by the transport mechanism**
 - E.g., IPsec
- End-to-end authentication
 - Provided by the transport mechanism (e.g. IPsec)

PPTP – Point-to-Point Tunneling Protocol



Original Reference Scenario



Customer provisioned deployment mode

General Features

- Adopted by IETF (RFC 2637)
- Tunneling of PPP frames over packet switched networks
- Microsoft Encryption: MPPE
- Microsoft Authentication: MS CHAP
- PPTP Network Server (PNS)
 - Corporate (VPN) gateway
- PPTP Access Concentrator (PAC)
 - For provider provisioned deployment mode



PPTP Connections

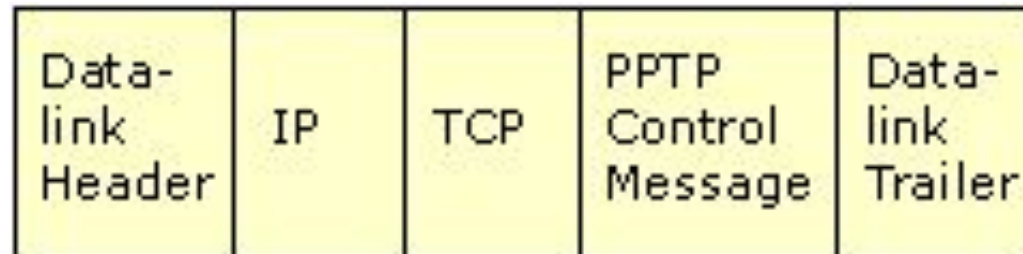
- PPTP Data Tunneling

- PPP tunneling
- GRE (of PPP over IP)



- Control Connection

- Data tunnel setup, management, and tear-down
- TCP encapsulation
 - PNS port 1723



PPTP Header



Value	Control Message
1	Start-Control-Connection-Request.
2	Start-Control-Connection-Reply.
3	Stop-Control-Connection-Request.
4	Stop-Control-Connection-Reply.
5	Echo-Request.
6	Echo-Reply.
7	Outgoing-Call-Request.
8	Outgoing-Call-Reply.
9	Incoming-Call-Request.
10	Incoming-Call-Reply.
11	Incoming-Call-Connected.
12	Call-Clear-Request.
13	Call-Disconnect-Notify.
14	WAN-Error-Notify.
15	Set-Link-Info.

IP sec VPN



Authentication Header Protocol (AH)

- Source authentication + data integrity
 - No confidentiality
- AH header inserted between IP header and payload
 - Protocol field: 51
- Routers process datagrams as usual
 - Not NAT, though



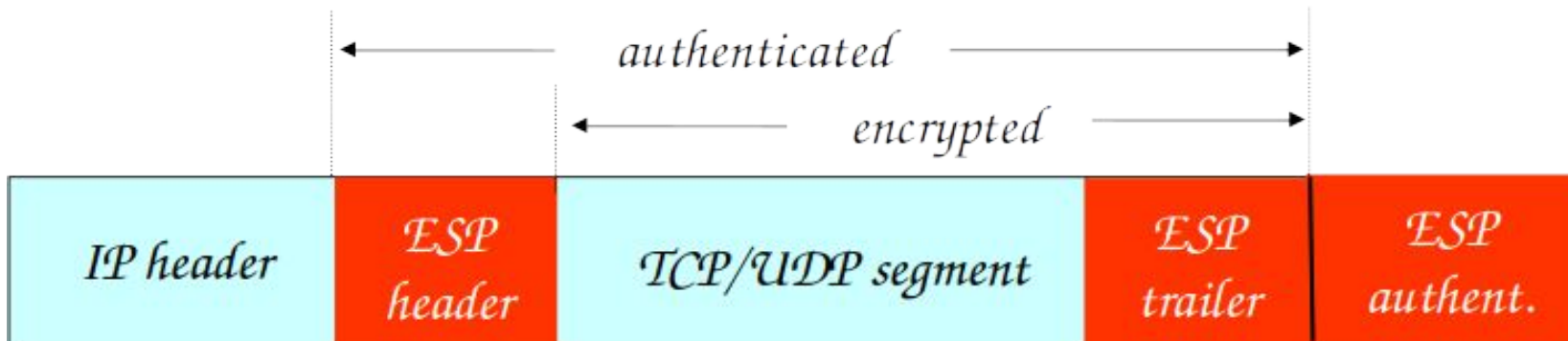
Authentication Header

- SPI: Security Parameter Index
 - Session ID
 - How to verify signature
 - Crypto algorithm
 - Reference to key
- Authentication data
 - Crypto signature
- Next header field
 - Protocol (e.g., TCP, UDP, ICMP) in payload



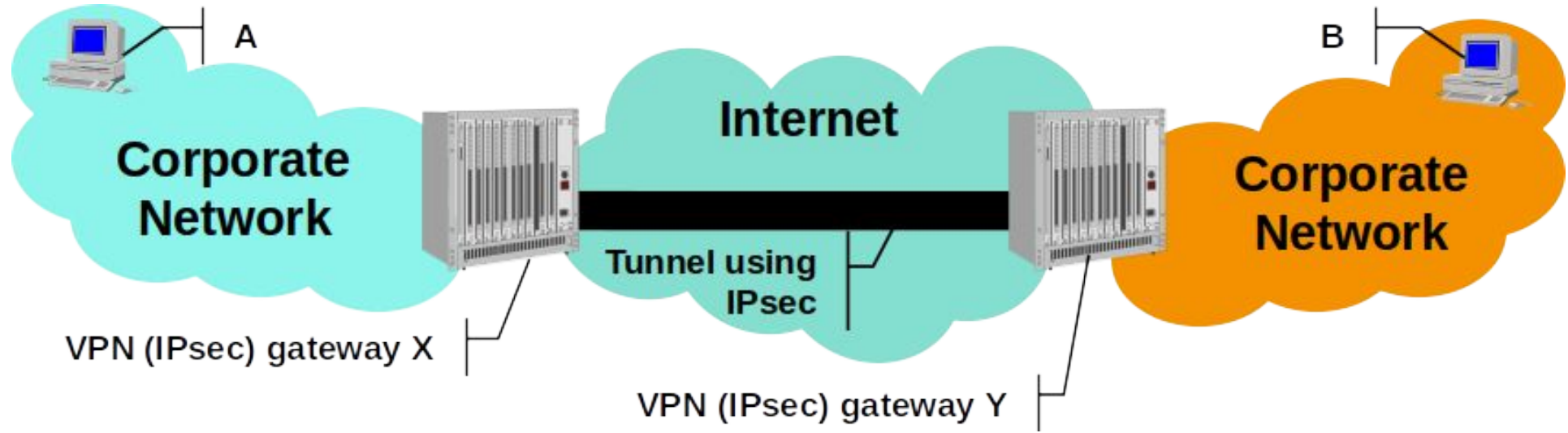
Encapsulating Security Payload (ESP)

- Data confidentiality
 - Data and ESP trailer encrypted
 - Next header field in ESP trailer
- Host authentication
- Data integrity
 - Authentication field similar to AH
- Protocol = 50



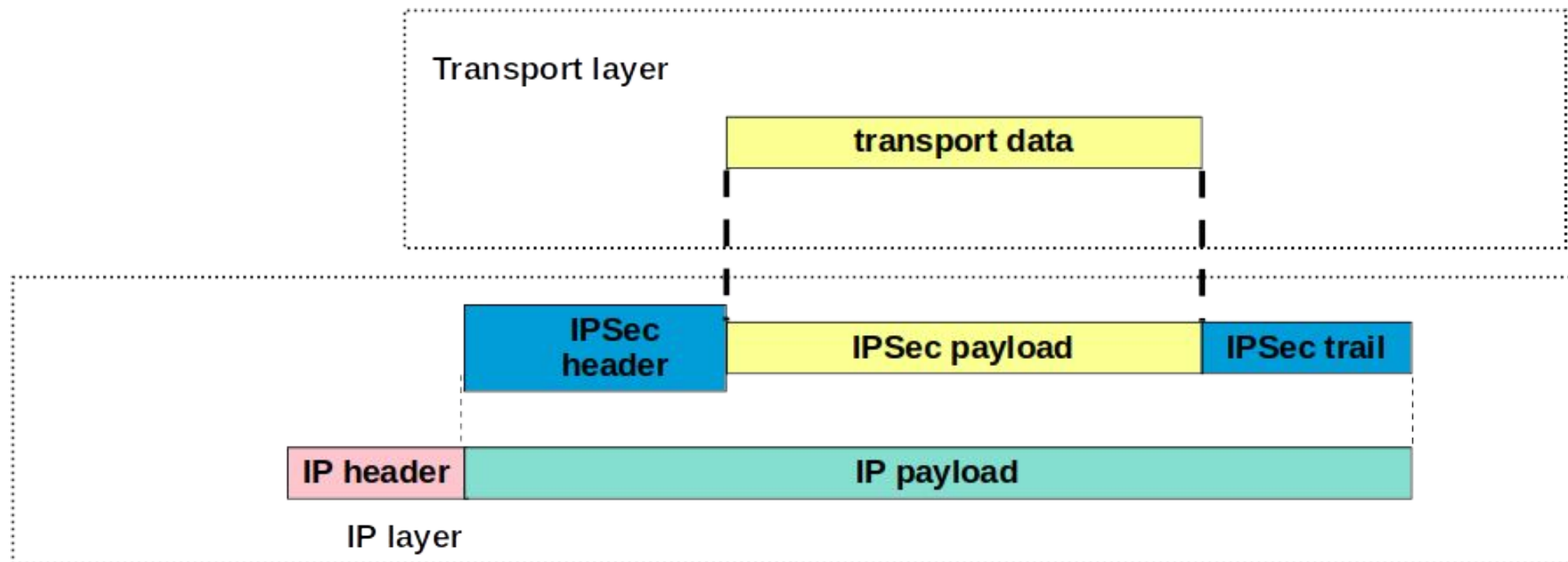
IPsec VPNs

- IPsec tunnel between VPN gateways
 - Encryption
 - Authentication
 - Encapsulation



Transport Mode

- IP header not fully protected
 - (only authenticated if AH is used)



Tunnel Mode

- It protects both IP header and payload

